Figure 2: Why DP-SGD Annihilates DRO — The Clipping Saturation Mechanism

(a) Clipping Saturation Theorem

In χ^2-DRO: $\tilde{g}_i = \text{Clip}(w_i^* \cdot g_i, C)$

Regime 1: Fairness Destruction

When $\|g_i\| \geq C$ and $w_i^* > 1$

$\text{Clip}(w_i^* g_i, C) = C\frac{g_i}{\|g_i\|}$

$= \text{Clip}(g_i, C)$

DRO weight cancels out!

Regime 2: Privacy Amplification

When $\|g_i\| < C < w_i^* \|g_i\|$

$\|\tilde{g}_i\| = C$ (ceiling)

Minority samples hit 100% of C
Majority samples ≈ 30% of C

Empirical Verification (Adult, ε=8, C=1.0)

Subgroup	n	E[w*]	Saturation	Post-Clip w	ΔLiRA TPR
White	38,903	0.82	11.4%	0.89×	+0.2%
Black	4,228	1.94	68.2%	1.41×	+1.8%
Asian-Pac-Isl	1,303	2.45	79.5%	1.35×	+2.6%
Amer-Ind-Esk	435	3.80	91.0%	1.12×	+3.9%
Other	353	5.12	96.8%	1.04×	+4.5%

(b) Gradient Norms per Subgroup (UTKFace, ε=8)

